

Sampling & Monte Carlo Methods

Lecture Notes (After Midterm)

ASM Course

Week08-10

Topics: Numerical Integration · Monte Carlo Simulation · Uniform RNG · Inverse Transform ·
Normal Generation · Multivariate Normal · Accept-Reject · Importance Sampling

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Chapter 1

Sampling and Monte Carlo Methods

1.1 Motivation: Numerical Integration

Sampling techniques: generate random variables from a distribution, as our end goal is to integrate (find expectations).

$$\mathbb{E}[g(X)] = \int_{\mathcal{X}} g(x) f_X(x) dx, \quad f_X : \text{density of } X.$$

Remark

Frequentist inference. Often we want to find the variance of an estimator (we need Fisher information), i.e. $\mathbb{E}[\text{second derivative of log-likelihood}]$.

Bayesian inference. $\mathbb{E}[\text{parameter under posterior distribution}]$.

There are ways of doing integration numerically. But why do these with random variables?

$$g : D \rightarrow \mathbb{R}, \quad D \subseteq \mathbb{R}^n, \quad I = \int_D g(x) dx.$$

If $D = 1$, **quadrature methods** can be used to approximate the integral. If m is the number of sub-intervals, then the error rate is $O(m^{-k})$, where k depends on the method used.

Example: Simpson's Rule

If Simpson's Method is used, then k is the number of nodes.

1.1.1 Drawbacks of Numerical Integration

- (1) We know the rate, but we do *not* have actual bounds on the errors:

$$\hat{I} - I = O(m^{-k}).$$

Do not know the constant in $O(\cdot)$.

- (2) For n dimensions, the order is $m^{-k/n}$. Needs a lot of computations - m^n to get the same level of accuracy. (A grid with m points per dimension in n dimensions requires m^n total evaluations - *curse of dimensionality*.)

What are we going to do to overcome these? **Sampling / Simulation / Monte Carlo methods.**

1.2 Monte Carlo / Simulation / Sampling Methods

Generate random variables

$$X_1, \dots, X_m \stackrel{\text{i.i.d.}}{\sim} U(D).$$

(1) (LLN)

$$\bar{g}_m(x) = \frac{1}{m} \sum_{i=1}^m g(X_i) \xrightarrow{a.s.} \frac{\int_D g(x) dx}{\text{area}(D)}.$$

(2) (CLT)

$$\sqrt{m} \frac{\bar{g}_m(x) - \frac{\int_D g(x) dx}{\text{area}(D)}}{\sqrt{\text{Var}(g(X))}} \Rightarrow N(0, 1).$$

These results do *not* depend on the dimension of D . (Area is replaced by volume of D in higher dimensions.)

The estimator of I and its error are:

$$\hat{I}_1 = \bar{g}(x) \cdot \text{area}(D), \quad \hat{I}_1 - I = \frac{\sqrt{\text{Var}}}{\sqrt{m}} \cdot \text{area}(D) \cdot Z, \quad Z \sim N(0, 1).$$

So we have a good idea of the bound on the error (it is a normal random variable).

Example: Estimating $\int_0^1 \cos x dx$

Let $g(x) = \cos x$, $D = [0, 1]$. Then $I = \sin 1 = 0.841471$.

m	$\hat{I} = \bar{g}_m(x)$
1	0.774
10	0.872
100	0.834
1000	0.840
10^5	0.841

1.3 Generating Uniform(0,1) Random Variables

Generating a $U(0, 1)$ sequence. Built-in R function $\rightarrow$ `runif`.

Requirements:

1. Each observation should be $U(0, 1)$.
2. Independent.

1.3.1 Pseudo-Random Numbers

Real random variable generation can be done by physical processes (e.g. coin flip). In practice we use **pseudo-random numbers**.

Linear Congruential Generator

Start with seed $x_0 \rightarrow$ integer. Fix integers a and m :

$$x_1 = a x_0 \pmod{m}, \quad x_2 = a x_1 \pmod{m}, \quad \dots$$

so $x_i \in \{0, 1, 2, \dots, m\}$ and

$$u_i = \frac{x_i}{m}.$$

$u_1, u_2, \dots$ is the output.

Note

m should be large. Avoid x_i being 0; m to be prime. If $a = 2$, then the cycle is of length 8. **Caution:** If you take too many random variables there will be repetitions.

Clear dependence between (u_i, u_{i+1}) , $i = 1, \dots, n$. (*Lattice structures.*)

1.3.2 Combining Generators

Several linear congruential generators with parameters $(a_1, m_1), \dots, (a_k, m_k)$, producing sequences $x_{1i}, \dots, x_{ki}$ and $u_{1i}, \dots, u_{ki}$. Set:

$$m = \max\{m_1, \dots, m_k\}, \quad Z_i = \sum_{l=1}^k (-1)^l x_{li},$$

$$u_i = \begin{cases} Z_i/m & \text{if } Z_i > 0, \\ \text{no output} & \text{otherwise.} \end{cases}$$

Alternative combination: $u_i = \left\{ \sum_{l=1}^k u_{li} \right\}$ (fractional part).

1.4 Generating General (Non-Uniform) Random Variables

Many standard distributions have built-in R functions to generate them (`rbinom`, `rexp`, etc).

If you want to generate from a **non-standard distribution**, one general procedure is to use the **Inverse CDF** method.

1.4.1 Inverse Transform Method

Let $X \sim F_X$, so $P(X \leq x) = F_X(x)$. Define:

$$F_X^{-1}(u) = \inf\{t : F_X(t) \geq u\}, \quad 0 \leq u \leq 1.$$

Theorem: Inverse Transform

$F_X(x) \geq u$ if and only if $F_X^{-1}(u) \leq x$. (i)

If $U \sim \text{Unif}(0, 1)$, set $X = F^{-1}(U)$. Then

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(F(x) \geq U) = F(x),$$

using (i) and the fact that $U \sim \text{Unif}$.

Example: Exponential distribution

$X \sim \text{Exp}(1)$: $F(x) = 1 - e^{-x}$. Setting $F(x) = u$:

$$F^{-1}(u) = -\ln(1 - u).$$

KolmogorovSmirnov (KS) Test Connection

Given $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$, the one-sample KS statistic is:

$$\sup_x |F(x) - \hat{F}_n(x)|.$$

Two-sample KS: $X_1, \dots, X_n \sim \text{Exp } F_X$ using built-in `rexp`; $Y_1, \dots, Y_m \sim \text{Exp } F_Y$ using inverse CDF. Statistic:

$$\sup_{0 \leq t \leq 1} |\hat{F}_X(t) - \hat{F}_Y(t)|.$$

Inversion for Standard Distributions

Logistic distribution.

$$F(x) = \frac{1}{1 + e^{-(x-\mu)/\lambda}} = u \implies F^{-1}(u) = -\lambda \ln\left(\frac{1}{u} - 1\right) + \mu.$$

Cauchy distribution.

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x - \mu}{\sigma}\right).$$

Use $\text{Exp}(\cdot)$ to generate $\text{Gamma}(\alpha, \beta)$, where α is an integer: sum of α many independent exponentials.

We cannot get closed form inverse CDF of Gamma , $\chi_{(2n)}^2$, $\text{Beta}(n_1, n_2)$ in general. We need other procedures.

1.4.2 Discrete Distributions

$$X = \begin{cases} x_1, & p_1 \\ \vdots & \\ x_k, & p_k \end{cases}, \quad \sum p_i = 1, p_i \geq 0, x_1 \leq x_2 \leq \cdots \leq x_k.$$

Use $U \sim \text{Unif}(0, 1)$. Compute $F(x_i) = \sum_{j=1}^i p_j$. Find i such that

$$F(x_{i-1}) < u \leq F(x_i), \quad \text{then set } X = x_i.$$

1.5 Generating Multinomial via Conditional Distributions

Let $\mathbf{X} = (X_1, \dots, X_k)^T$, $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$, $\mathbf{p} = (p_1, \dots, p_k)^T$, $p_i \geq 0$, $\sum p_i = 1$, $x_i \geq 0$, $\sum x_i = n$.

Remark

Inverse transform for multivariate CDF is not well-defined. If $U \sim U(0, 1)$ gives $u = 0.4$, then $F(\mathbf{x}) = 0.4$ has multiple solutions $\mathbf{x}$.

n independent trials. Each trial can lead to k outcomes. The j -th outcome happens with probability p_j .

If $\sum_{j=1}^{i-1} p_j < u < \sum_{j=1}^i p_j$, the outcome of a single trial is $(0, 0, \dots, \underbrace{1}_{\text{ith}}, 0, \dots, 0)$.

This is inefficient as it requires n many uniforms.

Alternatively, use conditional distributions. For $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$, $\mathbf{x} = (x_1, \dots, x_k)^T$:

- Marginal distribution of X_1 : $X_1 \sim \text{Bin}(n, p_1)$.
- $X_2 \mid X_1 \sim \text{Bin}\left(n - X_1, \frac{p_2}{p_2 + \cdots + p_k}\right)$.
- $\begin{pmatrix} X_2 \\ \vdots \\ X_k \end{pmatrix} \mid X_1 \sim \text{Mult}\left(n - X_1, \frac{(p_2, \dots, p_k)^T}{\sum_{i \geq 2} p_i}\right)$.
- $X_3 \mid (X_2, X_1) \sim \text{Bin}\left(n - X_1 - X_2, \frac{p_3}{p_3 + \cdots + p_k}\right)$.
- ... and $X_k = n - \sum_{i=1}^{k-1} X_i$.

Generate binomials by inverse transform. Only $(k - 1)$ uniforms needed.

1.6 Generating Univariate Normal

$X \sim N(\mu, \sigma^2)$: if $Z \sim N(0, 1)$ then $X = \mu + \sigma Z$. It is enough to develop methods for $N(0, 1)$.

1.6.1 Inverse Transform Approximation

$F \rightarrow F^{-1}$: there is no closed form for $F = \Phi$.

Approximation:

$$\Phi(x) \approx 1 - \phi(x)(b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5), \quad t = \frac{1}{1 + px},$$

with maximum absolute error 7.5×10^{-8} . The constants $b_1, b_2, \dots, b_5$ and p are known constants.

$\Phi(-x) = 1 - \Phi(x)$ for $x < 0$.

Inverse of Φ^* : solve $\Phi^*(x) - u = 0$ numerically using NewtonRaphson.

In R, `rnorm` by default uses $\Phi^*(\cdot)$. (Lots of approximations and takes a long time.)

1.6.2 BoxMuller Method

Generates $N(0, 1)$ in pairs. Need two uniforms. $R \sim \text{Exp}(2) \perp \Theta \sim U(0, 2\pi)$.

Procedure: BoxMuller Algorithm

1. Generate $U_1, U_2 \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1)$.
 2. Set $R = -2 \log U_1$ and $\Theta = 2\pi U_2$.
 3. Set $X_1 = \sqrt{R} \cos \Theta$ and $X_2 = \sqrt{R} \sin \Theta$.
- Then $X_1, X_2 \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$.

Verification. The joint density of (R, Θ) , $R \sim \text{Exp}(2)$, $\Theta \sim U(0, 2\pi)$, independently:

$$f(R, \Theta) = \frac{1}{2} e^{-R/2} \cdot \frac{1}{2\pi}, \quad R > 0, \quad 0 < \Theta < 2\pi.$$

$(X_1, X_2) \in \mathbb{R}^2$. The Jacobian of the inverse transform:

$$|J| = \left| \begin{array}{cc} \frac{\cos \Theta}{2\sqrt{R}} & \frac{\sin \Theta}{2\sqrt{R}} \\ -\sqrt{R} \sin \Theta & \sqrt{R} \cos \Theta \end{array} \right| = \frac{1}{2}.$$

So $|J^{-1}| = 2$, and:

$$f(x_1, x_2) = \frac{1}{2} \cdot \frac{1}{2\pi} \cdot 2 e^{-(x_1^2 + x_2^2)/2} = \frac{1}{\sqrt{2\pi}} e^{-x_1^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-x_2^2/2}.$$

So X_1 and X_2 are independent standard normals.

1.7 Generating Multivariate Normal

Multivariate Normal: $\mathbf{X} \sim N_d(\boldsymbol{\mu}, \Sigma)$, Σ p.d., $\Sigma = AA^T$.

If $\mathbf{Z} \sim N_d(\mathbf{0}, I)$, then

$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z} \sim N(\boldsymbol{\mu}, \Sigma).$$

$\mathbf{Z}$ is a vector of d many independent $N(0, 1)$'s. We know how to generate $N(0, 1)$.

1.7.1 Cholesky Decomposition

Cholesky decomposition gives A from Σ . $O(d^2/2)$ computations. Efficient. A is upper triangular.

1.7.2 Eigen-Decomposition / SVD

$$\Sigma = V\Lambda V^T,$$

where Λ is a diagonal matrix of eigenvalues of Σ , and V is the stack of eigenvectors.

Set $A = V\Lambda^{1/2}$; then $AA^T = V\Lambda V^T = \Sigma$.

Principal Components

Order eigenvalues in **decreasing** order. $\mathbf{X} = A\mathbf{Z} + \boldsymbol{\mu}$ implies:

$$\mathbf{Z} = A^{-1}(\mathbf{X} - \boldsymbol{\mu}) = \Lambda^{-1/2}V^T\mathbf{X}.$$

All components of $\mathbf{Z}$ are linear combinations of $\mathbf{X}$:

$$N(0, 1) \sim \mathbf{Z} = \Lambda^{-1/2}V^T\mathbf{X}, \quad Z_j = \lambda_j^{-1/2}\mathbf{v}_j^T\mathbf{X} \quad (d\text{-dimensional}).$$

Z_j are independent of each other. $\mathbf{v}_j^T\mathbf{X}$ are called **principal components** of $\mathbf{X}$:

$$\text{Var}(\mathbf{v}_j^T\mathbf{X}) = \lambda_j.$$

Chapter 2

Accept-Reject and Importance Sampling

2.1 Accept-Reject Method

There are many situations (normal, gamma, etc.) where we cannot use the inverse transform method to generate random variables, either because the CDF cannot be written in closed form or cannot be inverted analytically. There we need to talk about **indirect methods: Accept-Reject algorithm** → very general.

Want to generate $X \sim f$ (density). Find a different density g satisfying:

- (1) You can generate random variables from g .
- (2) f is “absolutely continuous” w.r.t. g : $g(E) = 0 \Rightarrow f(E) = 0$.

Remark

Absolutely continuous: If for any set E , $g(E) = 0$, then $f(E) = 0$. (f is also continuous w.r.t. g .)

Example: $g = U(0, 1)$, $f = \text{Gamma}(\cdot)$ will *not* work, as $g([3, 4]) = 0$ but $f([3, 4]) \neq 0$.

$g = \exp$, $f = \text{Gamma}(\cdot)$ will work.

f is called the **target density**; g is called the **candidate density**.

Definition: Target and Candidate Densities

- f : target density.
- g : candidate density (can generate Y from g ; f abs. continuous w.r.t. g).
- Require a constant M such that

$$M \geq \sup_{y \in \text{supp}(g)} \frac{f(y)}{g(y)}.$$

Procedure: Accept-Reject Algorithm

To simulate $X \sim f$:

1. Generate $Y \sim g$ and $U \sim \text{Unif}(0, 1)$ independently.
2. If

$$U \leq \frac{1}{M} \cdot \frac{f(Y)}{g(Y)},$$

set $X = Y$ (accept). Else discard Y and U ; go back to Step 1.

(Claim: $X \sim f$.)

Proof of claim.

$$P(Y \leq x \mid \text{Accept}) = P\left(Y \leq x \mid U \leq \frac{1}{M} \frac{f(Y)}{g(Y)}\right).$$

$$\text{LHS} = \frac{P\left(Y \leq x, U \leq \frac{1}{M} \frac{f(Y)}{g(Y)}\right)}{P\left(U \leq \frac{1}{M} \frac{f(Y)}{g(Y)}\right)}.$$

For the numerator:

$$\begin{aligned} P\left(Y \leq x, U \leq \frac{f(Y)}{Mg(Y)}\right) &= \int_{-\infty}^x P\left(U \leq \frac{f(y)}{Mg(y)}\right) g(y) dy \\ &= \int_{-\infty}^x \frac{f(y)}{Mg(y)} \cdot g(y) dy \quad [f(y) \neq 0 \Rightarrow g(y) \neq 0] \\ &= \frac{1}{M} \int_{-\infty}^x f(y) dy = \frac{1}{M} P(X \leq x). \end{aligned}$$

For the denominator:

$$P\left(U \leq \frac{f(Y)}{Mg(Y)}\right) = \int_{-\infty}^{\infty} \frac{f(y)}{Mg(y)} \cdot g(y) dy = \frac{1}{M} \int_{-\infty}^{\infty} f(y) dy = \frac{1}{M}.$$

Hence proved. □

Remark

- **Proportion of samples that are accepted** is $1/M$.
- Taking $M = \sup_{y \in \text{supp}(g)} f(y)/g(y)$ leads to least number of rejections. So we also want to choose g so that M is low.
- Accept-Reject method is applicable in any dimension.

Example: Generating from $\text{Beta}(a, b)$ via Accept-Reject (R-code)

Target: $X \sim \text{Beta}(a, b)$. Beta CDF is not available in closed form:

$$F_X(u) = \left(\int_0^u x^{a-1} (1-x)^{b-1} dx \right) / B(a, b), \quad 0 < u < 1.$$

Candidate: $Y \sim U(0, 1)$.

Importance Sampling

Evaluate integral $\rightarrow$ probability or expectation of a r.v.

1. Probability

$$\int_{-\infty}^k f(x) dx$$

$$\int_{-\infty}^{\infty} h(x)f(x) dx$$

$$h(x) = I(x \leq u)$$

2. Expectation of a RV

$$\int_{-\infty}^{\infty} xf(x) dx$$

3. Expectation of a function of X :

$$\int_{-\infty}^{\infty} h(x)f(x) dx$$

- 4.

$$\int g(x) dx = \int \frac{g(x)}{f(x)} f(x) dx$$

where f is a density of a r.v.

$$\int_{-\infty}^{\infty} h(x)f(x) dx$$

is the general form.

Objective is to find

$$\theta = \int h(x)f(x) dx$$

where f is a density.

$$= \mathbb{E}_f[h(X)] \quad (X \sim f)$$

So far we were generating $X_1, \dots, X_n$ from f . Then we can approximate / estimate the integral by (LLN):

$$\frac{1}{n} \sum_{i=1}^n h(X_i) = \hat{\theta}_f$$

$$\frac{1}{n} \sum_{i=1}^n h(x_i) = \hat{\theta}_n$$

$\hat{\theta}$ is consistent for θ by LLN.

Want to evaluate the integral

Difficult or impossible to generate $x_1, \dots, x_n$ from f .

Using “accept-reject” we can generate $x_1, \dots, x_n$ from f indirectly using g . But we lose a lot of samples (y). Not very efficient.

[In eg 10,000 random numbers were necessary to get $\sim 1250 = n$]

In the importance sampling approach we do not lose random numbers (more efficient), but we do not get x_i ’s explicitly. We get $\hat{\theta}$ directly from $Y_1, \dots, Y_n$.

$f \rightarrow$ target density

$g \rightarrow$ candidate density

[can generate y from g and f is abs continuous]

Objective is to estimate

$$\theta = \int h(x)f(x)dx \quad \text{for some } h.$$

$$\int h(x)f(x)dx = \int h(x)\frac{f(x)}{g(x)}g(x)dx$$

$$= \mathbb{E} \left(h(y)\frac{f(y)}{g(y)} \right) \quad \text{where } y \sim g$$

$$\hat{\theta}_g = \frac{1}{n} \sum_{i=1}^n h(y_i)\frac{f(y_i)}{g(y_i)} \rightarrow \theta \text{ by LLN}$$

where $Y_1, \dots, Y_n$ iid $\sim g$.

$$w(x) = \frac{f(x)}{g(x)} \quad \text{is called the “importance function”}.$$

Instead of finding average of $h(x_i)$ we are finding the (weighted) average of $h(y_i)w(y_i)$.

Objectives

1. Cannot generate from f .
2. Maybe able to reduce the variance of $\hat{\theta}$.

Proper choice of g may give

$$\text{Var}(\hat{\theta}_g) < \text{Var}(\hat{\theta}_f)$$

3. If f is known only upto a constant then one can use importance sampling.

4. Finding probabilities of rare events.

eg 4:

$$X \sim N(0, 1), \quad P(X < -5) = \theta$$

$$X_1, \dots, X_n$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq -5)$$

A hit is obtained once in 3 million (n) samples.

$$f \ll g \quad (\text{absolutely continuous}) : \text{ If } g(x) = 0 \text{ then } f(x) = 0$$

$$g \ll f \text{ is not required.}$$

There can be $x = x_0$ such that $f(x_0) = 0$ but $g(x_0) \neq 0$.

Points that are not in the support of f will not contribute to the sum as

$$\frac{f(x)}{g(x)} = 0$$

Good to have $g \ll f$ (then you don't waste any sampling points) but not necessary.

eg 4:

$$X \sim N(0, 2) = f, \quad P(X < -5) = \theta$$

$$h(x) = \mathbf{1}(x < -5)$$

$$\int h(x)f(x) dx = \int \mathbf{1}(x < -5)f(x) dx = E_f[\mathbf{1}(X < -5)] = P(X < -5)$$

$$Y \sim N(-5, 2) = g$$

$$P(Y < -5) = \frac{1}{2}$$

$$\frac{1}{N} \sum_{i=1}^N h(Y_i) \frac{f(Y_i)}{g(Y_i)} = \hat{\theta}$$

$$f \ll g$$

Say $X \sim f$. Given x_0 , importance sampling will give $\hat{P}(X < x_0)$.

Accept-reject will also give $\hat{P}(X < x_0)$.

Given α , if you want x_0 at $P(X \leq x_0) = \alpha$

Cannot get x_0 from importance sampling.

Accept-reject will give $X_1, \dots, X_n \sim f$

Find x_0 such $n\alpha$ many are less than x_0 .

$$\hat{x}_0 = x_{[n\alpha]}$$

Reduce variance of an integral approximation

$$\theta = \int h(x)f(x) dx$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n h(X_i), \quad X_1, \dots, X_n \text{ i.i.d } f$$

$$E(\hat{\theta}) = \theta \quad \hat{\theta} \text{ is consistent}$$

$$\text{Var}(\hat{\theta}) = \frac{1}{n} \text{Var}(h(X)) = \frac{1}{n} \left[\int h^2(x)f(x) dx - \theta^2 \right]$$

$$\tilde{\theta} = \frac{1}{n} \sum h(Y_i) \frac{f(Y_i)}{g(Y_i)}, \quad Y_1, \dots, Y_n \text{ i.i.d } g$$

$$E(\tilde{\theta}) = \theta \quad \text{and } \tilde{\theta} \text{ is consistent}$$

$$\text{Var}(\tilde{\theta}) = \frac{1}{n} \text{Var}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right] = \frac{1}{n} \left[\int h^2(y) \frac{f^2(y)}{g(y)} dy - \theta^2 \right]$$

$$n \left[\text{Var}(\hat{\theta}) - \text{Var}(\tilde{\theta}) \right] = \int h^2(y)f(y) dy - \int h^2(y) \frac{f^2(y)}{g(y)} dy$$

$$= \int \left[h^2(y)f(y) \left(1 - \frac{f(y)}{g(y)} \right) \right] dy \quad \text{to be large}$$

$$\frac{f(y)}{g(y)} < 1 \iff f(y) < g(y) \quad \text{not possible as both integrate to 1}$$

$$\frac{f(y)}{g(y)} \approx 1 \iff g \approx f$$

$$\frac{f(y)}{g(y)} \ll 1 \iff g > f$$

g large for large values of h^2

$$\frac{f(y)}{g(y)} \approx \text{const for } x < 5$$

eg: Integral approximation

$$\theta = \int_0^{10} \exp(-2|x-5|) dx$$

$$X - 5 \sim \text{DoubleExp}(2)$$

$$h(x) = 10 \exp(-2|x-5|) \quad f(x) = \frac{1}{10} \mathbf{1}(0 < x < 10)$$

$$\theta = \int h(x)f(x) dx = E_f[h(X)]$$

Using this f , we are getting a lot of X_i from regions where h is small.

$$g(x) \sim \mathcal{N}(5, 1)$$

eg Estimation of expectation when the target density is known upto a constant

$$X \sim f \quad f(x) = \frac{f_u(x)}{C_f}, \quad f_u(x) \text{ is known but } C_f \text{ not known}$$

$$\theta = \int h(x)f(x)dx$$

g candidate density

$$Y_1, \dots, Y_n \sim g$$

(1) We can generate $Y \sim g$

(2) $f \ll g$

$$g(y) = \frac{g_u(y)}{C_g}$$

G_u unknown C_g may or may not be known

$$\hat{\theta} = \frac{1}{n} \sum h(y_i) \frac{f(y_i)}{g(y_i)} = \frac{1}{n} \sum h(y_i) \frac{f_u(y_i)}{g_u(y_i)} \cdot \frac{C_g}{C_f}$$

$$\frac{C_f}{C_g} = \phi \quad (\text{unknown parameter})$$

$$\phi = \frac{C_f}{C_g} = \frac{\int f_u(x) dx}{C_g} = \int \frac{f_u(x)}{g_u(x)} \frac{g_u(x)}{C_g} dx = \mathbb{E}_g \left[\frac{f_u(y)}{g_u(y)} \right]$$

$$\hat{\phi} = \frac{1}{n} \sum_{i=1}^n \frac{f_u(y_i)}{g_u(y_i)}$$

$$\tilde{\theta} = \frac{\frac{1}{n} \sum h(y_i) \frac{f_u(y_i)}{g_u(y_i)}}{\frac{1}{n} \sum \frac{f_u(y_i)}{g_u(y_i)}}$$

2.2 Summary: Accept-Reject vs. Importance Sampling

	Accept-Reject	Importance Sampling
Output	Exact samples from f	Weighted estimate of θ
Samples used	Only accepted ones	All samples (reweighted)
Requirement	Upper bound M on f/g	$f \ll g$ (no bound needed)
Works when f unnorm.	Yes	Yes (self-normalised)
Scales with dim.	Poorly (M grows)	Better in practice
Quantile estimation	Yes ($\hat{x}_0 = X_{(\lceil n\alpha \rceil)}$)	No (only gives $\hat{P}(X < x_0)$)